Xiaochuan Li

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Last updated: October 2026

Questions from linguistics and cognitive science, turned into controlled experiments on language models: behavioural tests, then mechanistic interpretability to locate the behaviour inside the model. Research interests: language, reasoning, and human and artificial cognition.

EDUCATION

University of Technology Nuremberg — Nuremberg, Germany
M.Sc. Human and Artificial Intelligence — *Oct. 2025 – exp. 2027*

Designing and analysing behavioural experiments; cognitive psychology; training neural networks; interpretability methods for language models.

University of Marburg — Marburg, Germany
German for graduate study — *2024 – 2025*

Shenzhen University — Shenzhen, China
B.A. Philosophy — *2020 – 2024*

Formal logic and proof, including modal logic and Gödel's incompleteness theorems; philosophy of language; psychology of reading.

RESEARCH

Moral framing and dilemma choices in language models — *2026–27*
Supervised at DFKI

Investigated how moral wording affects dilemma choices. Designed and ran controlled experiments and activation interventions in 7B and 14B models for the utilitarian-framing branch, with checks of answer labels, option positions, and task understanding; manuscript in preparation.

Property updating in modified nouns in language models — *2026–*
Independent project

Contrasted properties that depend on an added feature with properties that only co-occur with it (13 feature groups, 477 items, three instruction-tuned models), using predictors fixed outside the models: a dependency table, human feature norms and word-association norms; exploratory activation patching. Manuscript in preparation.

Modal language in preschool children — *2026 –*
M.Sc. thesis, design stage · part of a PhD project at the Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig

Asks how preschool children come to distinguish possibility from necessity in language, and how this relates to their nonlinguistic reasoning about possibilities; a behavioural study; data collection has not started. It lays the groundwork for later work on modal language in language models.

PROJECTS

Typicality in referent choice: what the description leaves open — *2026*
Independent theoretical study. Distinguishes three conditions for typicality in referent choice and the limits of what a forced-choice task can establish. Li, X. (2026). Working paper. lingbuzz/010343 · doi:10.5281/zenodo.22855036

Cross-lingual ProtoBias *Summer 2026*

Single-author course project. Evaluated two VLM families on fixed image pairs across seven prompt languages (about 12,600 judgments); single-author report with code. report · code

Street-view country classification *Summer 2026*

Team course project, Deep Learning. My part: CNN training, ablations, and performance analysis.

Fact-checking after ChatGPT exposure *Winter 2025–26*

Course study, Foundations in Psychology and Empirical Study Design. Between-subjects experiment on optional checking after ChatGPT or Wikipedia answers (49 participants, 47 analysed); co-designed the study, ran the data analysis and wrote the report.

Leibniz's logical system and characteristica universalis *2024*

Undergraduate thesis · Shenzhen University. Analysed Leibniz's metaphysics, logic and proposal for a universal symbolic language.

EMPLOYMENT

University of Technology Nuremberg Nuremberg, Germany

IT Service Desk (part-time) *Apr. 2026 – present*

Delivered student IT onboarding and individual support; handled system installation, service reporting, and inventory.

Shenzhen Learning Disorders Association Shenzhen, China

Internship · Learning assessment at Yucai No. 2 Primary School *Sep.–Dec. 2021*

Prepared character-reading materials, administered timed assessments, and analysed response data to support the assessment of learning difficulties; contributed to teaching items and individual learning plans.

SKILLS AND LANGUAGES

Language models	Controlled behavioural evaluation, minimal pairs; activation patching, low-rank analysis, probing, steering
Human data	Experimental design, PsychoPy/Pavlovia, JASP; testing children
Programming	Python (PyTorch, Hugging Face Transformers, pandas, scikit-learn), Git, LaTeX, Slurm on an HPC cluster
Languages	Mandarin Chinese (native), German (C1), English (C1), Japanese (learning)